SCHULICH School of Engineering

UNIVERSITY OF CALGARY

SENG 438
Software Testing, Reliability & Quality

Assignment 5

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- This assignment includes analysis of integration test data using reliability assessment tools
- There are two ways to assess failure data:
  - Part 1: Reliability growth testing
  - Part 2: Reliability Demonstration Chart (RDC)

Activities

- 1st week:
  - Install CASRE and/or SRTAT reliability growth tool
  - Install RDC tool (included in SRTAT or standalone EXCEL sheet)
  - Get failure data and try to run the tools using the data
- 2nd week:
  - Demo both parts

Assignment 5 – Part 1

- Run CASRE or SRTAT
- Import failure data
- Select a set of models that would provide the best fit for the project data using model ranking
- Select the range of useful data using Laplace or other tests
- Display time-between-failures, failure intensity and reliability graphs
- Discuss the acceptable range of failure rate for the test data

Note

- If none of the above mentioned tools work for you, try to find other tools that may work
- A list of those tools (freeware) are given below:
  - https://sasdlc.org/lab/assets/projects/srt.html
  - Software Defect Estimation Tool (SweET)
  - GITHUB: https://github.com/LanceFiondella/SwEET
  - Software Failure and Reliability Assessment Tool (SFRAT)
  - GITHUB: https://github.com/LanceFiondella/srt.core

Assignment 5 – Part 2

- Make yourself familiar with the RDC
- Input the failure data (failure number and failure time); identify the target MTTF and anticipated confidence levels, draw the failure points on the graph and analyze the trend
- Experiment with "what-if" scenarios by setting various values for MTTF and draw the plot
- Select the minimum $MTTF_{min}$ for which the SUT becomes acceptable
- Set MTTF to twice and half $MTTF_{min}$ and plot failure data
- Document the results